Practice workbook · Plan an investigation measuring a response or treatment in people

Pulse, reaction time, pupil size, catching, hearing, sweating or urine volume in volunteers, and treated versus untreated group trials of drugs or diets.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-PRAC-PLAN-INVESTIGATION-ON-PEOPLE

1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5 (a) A red blood cell travelling from the liver to the kidney passes through six different arteries and veins.

Complete the flow chart by naming these blood vessels in order. The first one has been done for you.

(5)

LIVER

↓

hepatic vein

↓

.....

↓

.....

↓

.....

↓

.....

↓

.....

↓

KIDNEY

OLevels.com SCHOOL

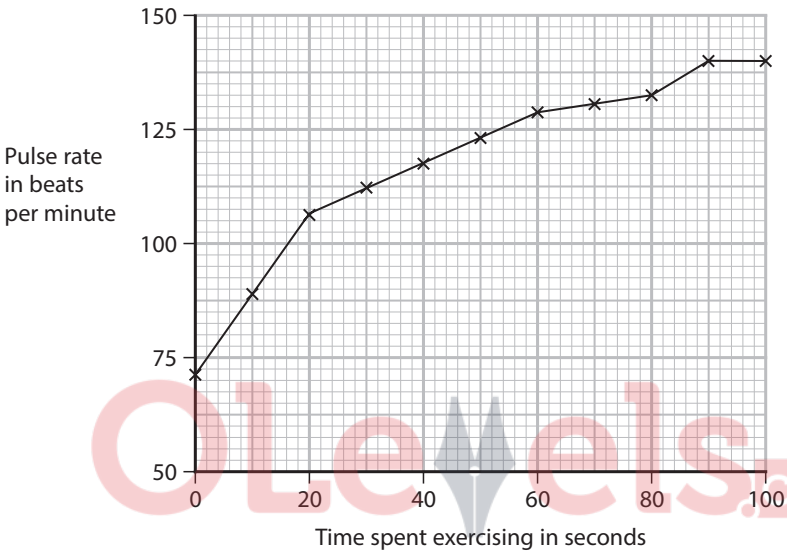
1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

(b) A student investigates the effect of time spent exercising on pulse rate.

The graph shows the student's results.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Explain the change in pulse rate recorded by the student.

(4)

OLevels.com SCHOOL

1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

(ii) Describe how the student could carry out the investigation.

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NO

OLevels.com SCHOOL

(Total for Question 5 = 15 marks)

1. Question 5 · part b.ii · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part b · 4 marks

January 2020 Paper 2

Further practice · Attempt part b only.

2 A student records her pulse rate in beats per minute (bpm) after different activities.

These are the student's results.

- at rest 78 bpm
- walking 95 bpm
- jogging 138 bpm
- running 146 bpm

(a) Draw a table to show the student's results.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 2 · part b · 4 marks

January 2020 Paper 2

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Describe a method that the student could use to obtain these results.

(4)

OLevels.com SCHOOL

(c) State one factor that the student should control during the investigation.

(1)

(d) Name a body process, other than pulse rate, that increases during exercise.

(1)

(Total for Question 2 = 9 marks)

2. Question 2 · part b · 4 marks · Working and answer

Working and answer

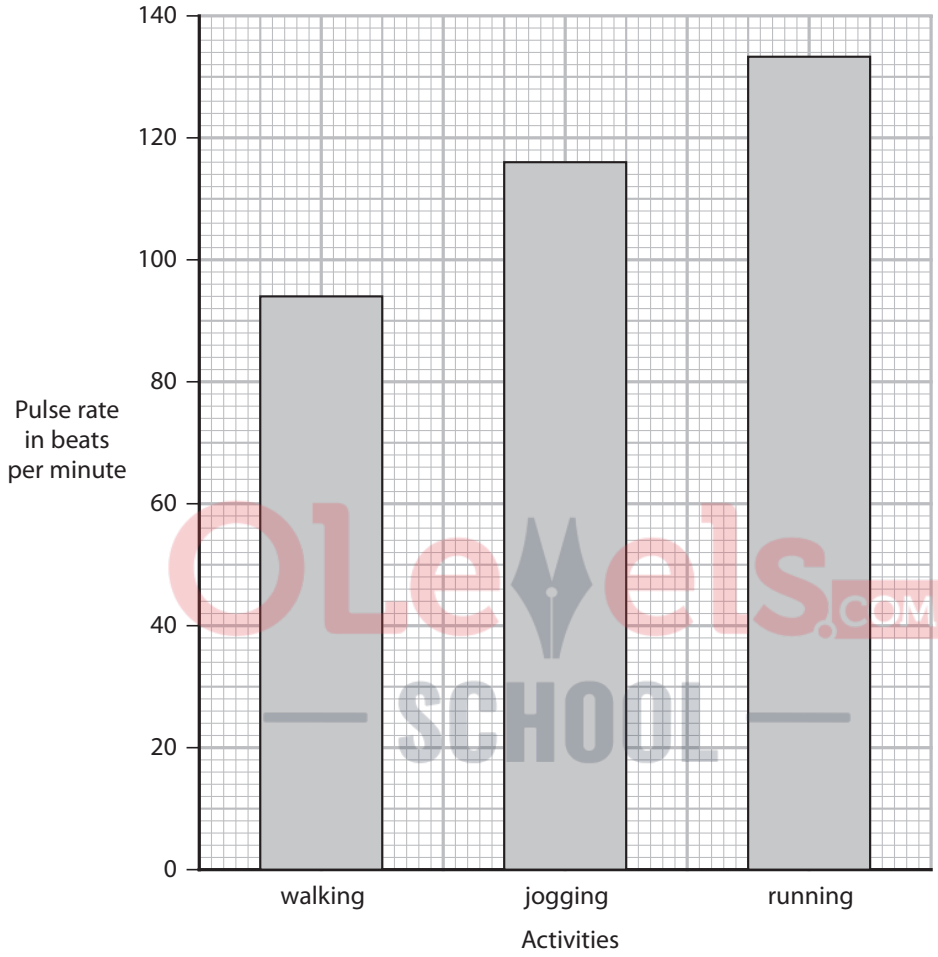
OLevels.com SCHOOL

3. Question 4 · part a.iii · 4 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

4 (a) The bar chart shows the pulse rate of a student at the end of three different activities.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 4 · part a.iii · 4 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Draw a table that the student could use to record these results.

(3)

(ii) Complete the table by adding the student's results.

(2)

(iii) Describe a method that the student could use to obtain the results shown in the bar chart.

(4)

OLevels.com — SCHOOL —

3. Question 4 · part a.iii · 4 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

(b) The pulse rates at the end of walking and running are different.

Explain this difference.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE I

OLevels.com SCHOOL

(Total for Question 4 = 13 marks)

3. Question 4 · part a.iii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 3 · part b.i · 6 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

OLevels.com SCHOOL

(ii) Describe the function of a motor neurone.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 3 · part b.i · 6 marks

June 2025 Paper 2R

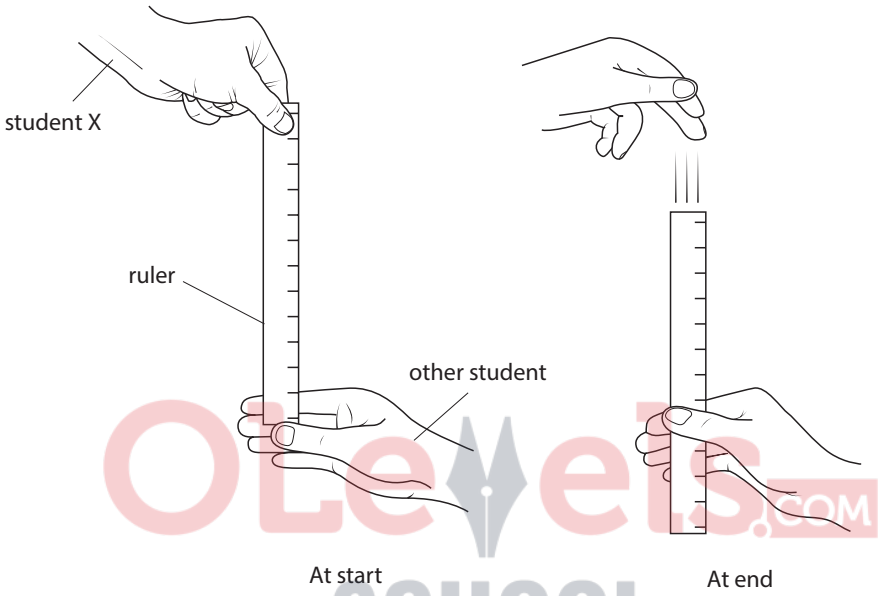
Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Student X uses this equipment to investigate the reaction time of another student.



Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

OLevels.com SCHOOL

4. Question 3 · part b.i · 6 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

(6)

(ii) Name the dependent variable in this investigation.

(1)

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

(1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 3 · part b.i · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part c.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

.....

2 .....

.....

(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

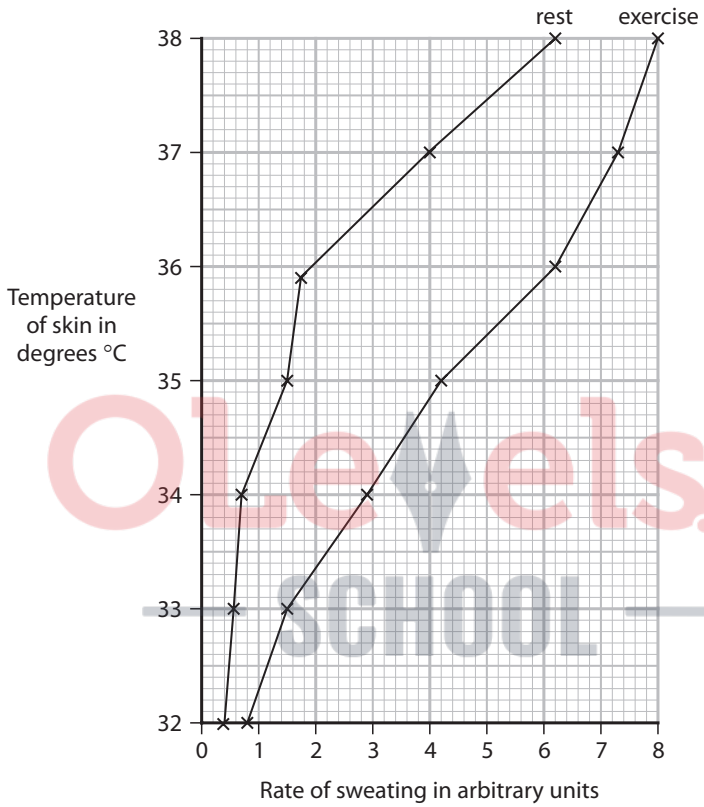
OLevels.com SCHOOL

5. Question 3 · part c.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.ii only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 3 · part c.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Describe how the skin temperature varies as the rate of sweating changes for the person at rest.

(2)

(ii) Suggest a method that could be used to measure the rate of sweating.

(2)

OLevels.com SCHOOL

(d) Explain the importance of maintaining a constant body temperature.

(3)

(Total for Question 3 = 13 marks)

5. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

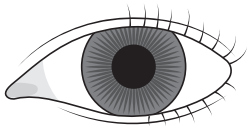
OLevels.com SCHOOL

6. Question 6 · part a · 3 marks

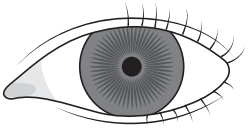
June 2024 Paper 2

Further practice · Attempt part a only.

6 The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

(3)

(b) A student calculated the area of the pupil in bright light to be 7.1 mm^2.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = πr^2 $\pi = 3.14$]

(3)

area of pupil in dim light = ... mm^2

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 6 · part a · 3 marks · Working and answer

Working and answer

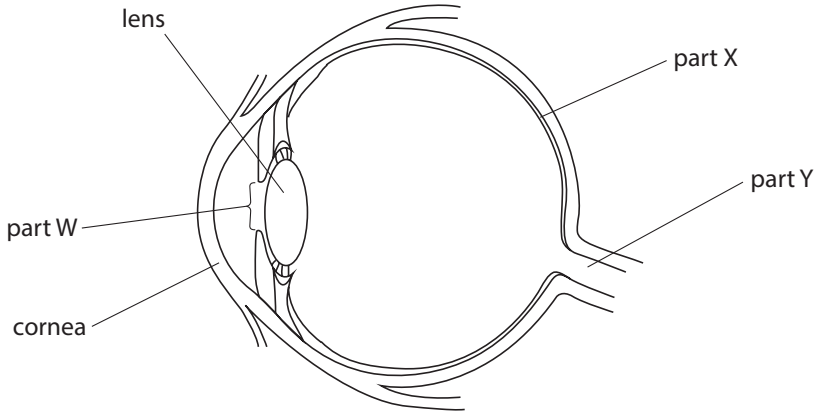
OLevels.com SCHOOL

7. Question 3 · part b · 4 marks

November 2024 Paper 1

Further practice · Attempt part b only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

.....

.....

.....

.....

(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 3 · part b · 4 marks

November 2024 Paper 1

Further practice · Attempt part b only.

(iv) Describe how corneal transplants help to restore normal vision for people with corneal disease.

(2)

(b) The iris reflex controls the amount of light entering the eye.

Describe a method that could be used to investigate how the eye responds to different light intensities.

(4)

(Total for Question 3 = 12 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 3 · part b · 4 marks · Working and answer

Working and answer

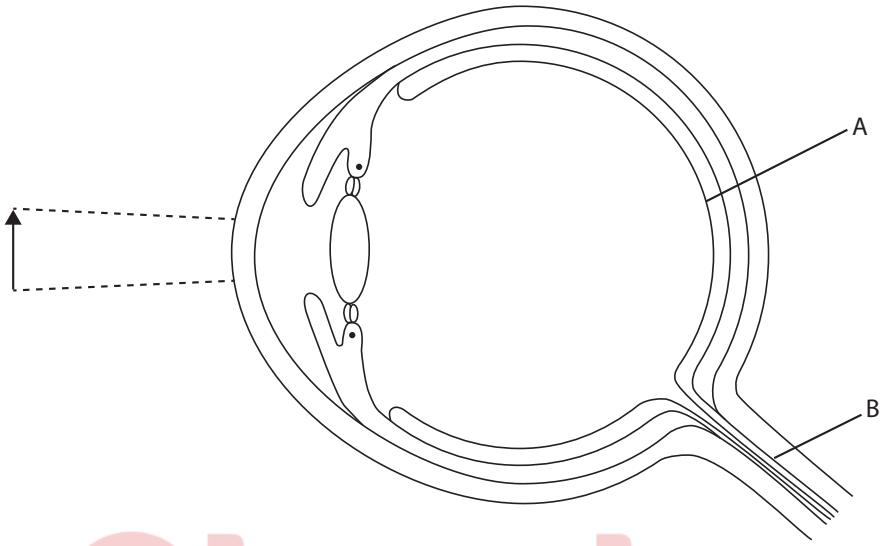
OLevels.com SCHOOL

8. Question 6 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

6 The diagram shows a section through the eye.



(a) (i) Describe the functions of structures A and B. **(4)**

A .....

.....

.....

.....

B .....

.....

.....

.....

(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8. Question 6 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Describe a simple experiment to show that stereoscopic vision would help a person to catch a ball thrown towards them.

(4)

(c) Describe what is meant by the term **astigmatism**.

(2)

(Total for Question 6 = 14 marks)

OLevels.COM — SCHOOL —

8. Question 6 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 5 · part a.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

9. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 5 · part a · 4 marks

June 2022 Paper 1

Further practice · Attempt part a only.

5 (a) There is a difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Design an investigation to determine the difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Include experimental details in your answer and write in full sentences.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 5 · part a · 4 marks

June 2022 Paper 1

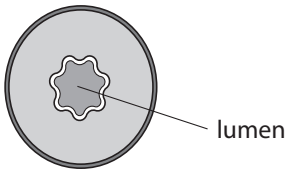
Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) The diagram shows a cross-section of a blood vessel in the skin on a cold day.



Explain the changes in this blood vessel on a hot day.

(3)

OLevels.com SCHOOL

(c) Describe how ADH regulates the volume of water in the body on a hot day.

(4)

(Total for Question 5 = 11 marks)

10. Question 5 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 4 · part d · 3 marks

June 2019 Paper 2

Further practice · Attempt part d only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

.....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

11. Question 4 · part d · 3 marks

June 2019 Paper 2

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(d) Describe a method that Goldberger could use to show that the symptoms of his patients were caused by a lack of vitamin B in their diet.

(3)

(e) Scurvy is another deficiency disease.

Scurvy affects the production of a protein called collagen.

Collagen is a component of many body tissues including cartilage.

Explain why a person with scurvy often experiences pain when walking.

(2)

(Total for Question 4 = 9 marks)

OLevels.COM SCHOOL

11. Question 4 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 9 · part a · 3 marks

June 2022 Paper 1R

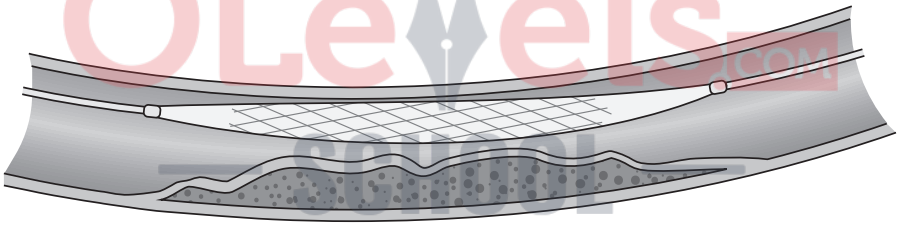
Further practice · Attempt part a only.

9 (a) Statins can be used to treat people with circulatory disorders.

Design a method that can be used to test the effectiveness of statins in treating circulatory disorders.

(3)

(b) The diagram shows a stent being used in the treatment of heart disease.



(Source: © DesignPrax/Shutterstock)

Explain how a stent can be used to treat heart disease.

(4)

(Total for Question 9 = 7 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN TH

12. Question 9 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 6 · part b.iii · 5 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

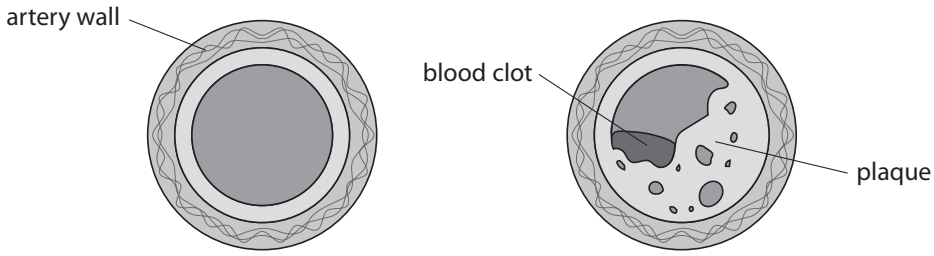
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

OLevels.com SCHOOL

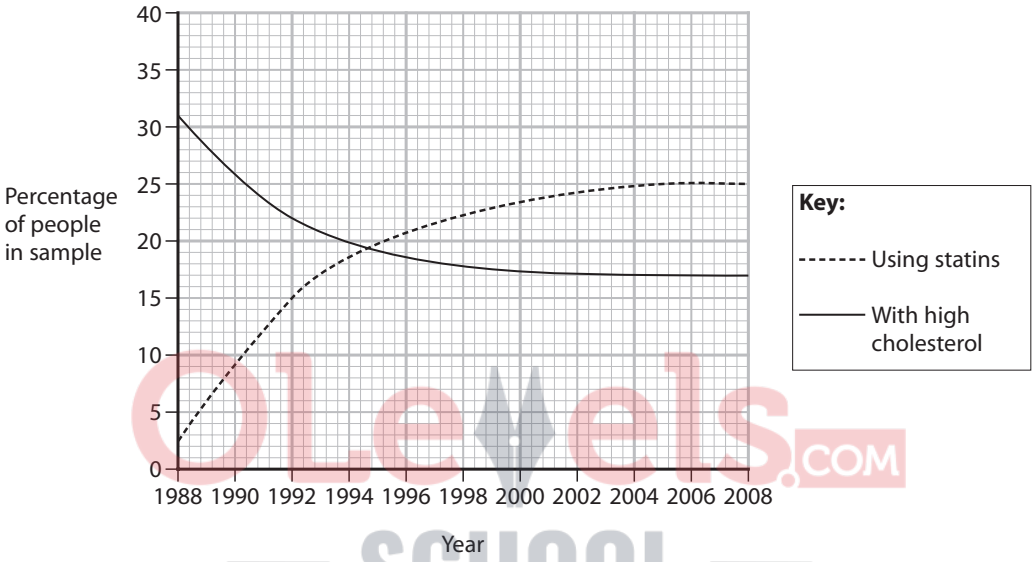
13. Question 6 · part b.iii · 5 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

13. Question 6 · part b.iii · 5 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

NOT WRITE IN THIS AREA

(iii) Describe a method that scientists could use to obtain the data shown in the graph.

(5)

OLevels.COM SCHOOL

(iv) Give two factors the scientists need to consider to make sure that their investigation is valid.

(2)

1 .....

.....

2 .....

.....

(Total for Question 6 = 15 marks)

13. Question 6 · part b.iii · 5 marks · Working and answer

Working and answer

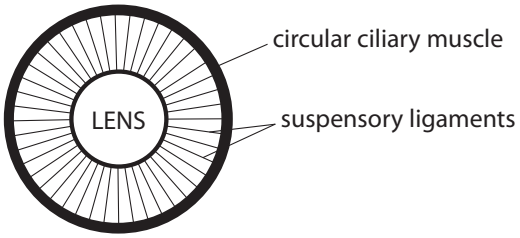
OLevels.com SCHOOL

14. Question 5 · part c · 6 marks

January 2022 Paper 2

Further practice · Attempt part c only.

5 (a) The diagram shows a front view of structures in the eye that help to form an image on the retina.



The three structures shown in the diagram change when focusing on near and distant objects.

Complete the table by putting ticks (✓) in the correct boxes to show the condition of each structure of the eye when focusing on near and distant objects.

(3)

	Focusing on a near object	Focusing on a distant object
Lens	thin ☐ thick ☐	thin ☐ thick ☐
Circular ciliary muscle	contracted ☐ relaxed ☐	contracted ☐ relaxed ☐
Suspensory ligament	stretched ☐ loose ☐	stretched ☐ loose ☐

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

14. Question 5 · part c · 6 marks

January 2022 Paper 2

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Humans have stereoscopic vision.

Explain what is meant by the term **stereoscopic vision**.

(3)

OLevels.com SCHOOL

14. Question 5 · part c · 6 marks

January 2022 Paper 2

Further practice · Attempt part c only.

(c) Design an investigation to show that it is easier for a person to catch a ball if they have stereoscopic vision.

Your answer should include experimental details and be written in complete sentences.

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 5 = 12 marks)

14. Question 5 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL